

Sprint Product Review – Sprint 1

Project: AI/ML Incident Response Optimization

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1. Sprint Goal and Accomplishments

The goal of Sprint 1 was to build a foundation for analyzing incident data so that stakeholders can better understand system performance and make data-driven decisions. This included tracking incident volume, measuring resolution time, and evaluating SLA compliance.

Overall, our team successfully met the sprint goal. We completed the main user stories that focused on data preparation and core performance metrics. Eric handled the data cleaning and validation, which was critical for ensuring the dataset was accurate and usable. After that, we were able to build key analytics, including resolution time metrics and SLA tracking.

The work completed in this sprint supports the overall project vision by providing initial insights into incident trends and performance. These results will be used in later sprints for deeper analysis and predictive modeling.

2. Software Demonstration

During Sprint 1, we implemented several key features using the ServiceNow incident dataset.

User Story 6: Data Cleaning and Validation (Eric)

Before performing any analysis, the dataset needed to be cleaned and standardized. The original data contained missing values, duplicate records, and inconsistent date formats, which would have affected the accuracy of any calculations.

This was handled by converting all timestamp fields into a consistent format, removing invalid or duplicate entries, and filtering out incorrect values such as negative resolution times and extreme outliers. The dataset was then aggregated at the incident level and exported as a cleaned file for further analysis.

This step was essential because all of the analytics in this sprint depend on having reliable and consistent data.

User Story 2: Average and Median Resolution Time (Latifa)

Using the cleaned dataset, the next step was to analyze how long it takes to resolve incidents.

Resolution time was calculated during the data preparation phase, so the focus of this user story was on analyzing those values. The data was grouped by priority level, and both the average (mean) and median resolution times were calculated.

Using both metrics provides a more complete view of performance. The average shows the overall trend, while the median helps reduce the effect of extreme values in the dataset.

A visualization was then created to compare resolution times across different priority levels. This makes it easier to see whether higher priority incidents are being handled more efficiently and helps identify any performance gaps.

This user story met the requirements by:

- using the cleaned dataset
- grouping incidents by priority
- calculating both average and median resolution time
- presenting the results in a clear and interpretable visualization

User Story 1: Incident Volume Dashboard

This user story was started during the sprint but was not fully completed. The goal was to create a dashboard showing incident volume over time to help identify trends and workload patterns.

Some initial work was done in preparing the data, but the final visualization was not completed within the sprint timeframe. This will be prioritized for completion in the next sprint.

User Story 3: SLA Adherence Tracking

This user story was partially addressed during the sprint. The objective was to calculate SLA compliance and identify violations.

While the team discussed the logic and structure needed for SLA tracking, the full implementation and visualization were not completed. This will be continued and finalized in the next sprint.

3. Challenges and Resolutions

One of the main challenges during this sprint was working with a messy dataset. The original data included missing values, inconsistent formats, and incorrect timestamps, which made it difficult to perform accurate calculations at the beginning.

This was addressed by focusing on data cleaning early in the sprint. Once the dataset was properly cleaned and validated, the analysis became much more straightforward.

Another challenge was managing time and completing all planned user stories within the sprint. While the team successfully completed data cleaning and resolution time analysis, some features, such as the incident volume dashboard and SLA tracking, were not fully completed.

To address this, the team plans to better estimate task complexity and prioritize workload more effectively in the next sprint.